

**MINISTRY OF HIGHER EDUCATION, SCIENCE AND
INNOVATIONS OF THE REPUBLIC OF UZBEKISTAN**

**NATIONAL QUALIFICATION REQUIREMENTS
FOR
60610300 –COMPUTER ENGINEERING
(BACHELOR)**

Tashkent -2024

DEVELOPED AND INTRODUCED BY:

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APPROVED AND PUT INTO EFFECT:

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PROPOSED:

Ministry of Higher Education, Science and Innovations of the Republic of
Uzbekistan:

These Qualification Requirements are based on the "State Educational Standard of Higher Education. General Provisions" and the "State Educational Standard of Higher Education. Classifier of Higher Education Fields and Specialties," the National and Sectoral Qualification Frameworks of the Republic of Uzbekistan, professional standards, and the proposals of personnel clients, and is an official normative-methodological document.

The right to officially publish Qualification Requirements in the territory of the Republic of Uzbekistan belongs to the Ministry of Higher Education, Science and Innovations of the Republic of Uzbekistan.

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1. GENERAL DESCRIPTION

The Bachelor's Degree Program in **60610300 – Computer Engineering** is offered in **full-time, evening, part-time, and distance learning modes**. Instruction in all modes of study is organized on the basis of the **credit-module system** (except for higher military educational institutions). The standard duration of the full-time bachelor's degree program is **four years**.

1.1 Scope of Application

1.1.1 Application of the Qualification Requirements

These qualification requirements establish a comprehensive set of standards for all higher education institutions that provide bachelor's degree training in the field of **60610300 – Computer Engineering**.

1.1.2. Main Users of the Qualification Requirements

The qualification requirements for this educational program serve the following stakeholders:

Administrative personnel of higher education institutions responsible for developing, updating, and effectively implementing qualification requirements, curricula, and academic programs, including rectors, vice-rectors, heads of academic departments, deans, and department chairs, as well as academic staff responsible for ensuring graduates achieve the required level of competence;

Students enrolled in higher education institutions who undertake the curriculum and academic programs of this field of study;

State Attestation Commissions responsible for assessing the qualification level and academic preparedness of bachelor's degree graduates;

Authorized state bodies responsible for educational governance and administration;

Organizations responsible for financing higher education institutions;

Authorized state bodies responsible for accreditation and quality assurance within the higher education system;

Employers, workforce customers, organizations, enterprises, and institutions requiring qualified specialists;

Applicants seeking admission to higher education institutions, their parents, and other interested stakeholders.

1.2. Description of Professional Activities

1.2.1 Areas of Professional Activity

The professional activity encompasses solving complex software engineering problems in the field of information and communication technologies in state and non-state enterprises, organizations, institutions, and public administration bodies, as well as participating in scientific research conducted by the Academy of Sciences of the Republic of Uzbekistan, sectoral research institutes, research centers, and scientific-production associations.

1.2.2 Objects of Professional Activity

Graduates of the **60610300 — Computer Engineering** bachelor's degree program may carry out professional activities in relation to:

- Mathematical and software support of automated and computer systems; computer-aided design systems;
- digital technologies and enterprises operating within the system of relevant ministries;
- manufacturing and service sectors; information processing and software support of microprocessor-based systems;
- effective processes for the development, implementation, and operation of software products and hardware components of computing equipment;

- processes of designing, configuring, manufacturing, and implementing technical and software tools of information and communication systems and technologies.

Graduates of the Bachelor's degree program **60610300 — Computer Engineering** shall, upon completion of pedagogical retraining, have the right to engage in pedagogical activities in preschool and school education organizations, as well as in professional educational institutions, by teaching general professional and specialized disciplines determined by the authorized educational governing bodies.

1.2.3 Types of Professional Activities

Graduates of the **60610300 — Computer Engineering** Bachelor's Degree Program may engage in the following types of professional activities:

- production activities;
- organizational and managerial activities;
- design and engineering activities;
- information and analytical activities;
- engineering and entrepreneurial activities;
- activities related to the application of multimedia technologies;
- activities related to ensuring information security of computer systems;
- activities related to intelligent data analysis.

1.2.4 Professional Functions and Responsibilities

In accordance with **Level 6 of the National Qualifications Framework**, and considering the professional fields, objects, and types of activities associated with the **60610300 — Computer Engineering** Bachelor's Degree Program, graduates are expected to possess the knowledge, skills, and competencies required to effectively perform professional functions in software engineering and information and communication technologies (ICT), including

software design, development, testing, deployment, maintenance, project management, and research activities.

Organizational and management activities

Within the framework of organizational and management activities, graduates shall be able to:

- develop methods and mechanisms for monitoring and assessing the quality of production processes related to software development and operation;
- monitor production processes for compliance with environmental protection and occupational safety requirements;
- develop and implement quality management processes for software production;
- manage quality in the production process;
- organize and manage the work of the team of performers;
- organize and manage the work of primary design, technological, or production teams;
- develop, implement, monitor, and evaluate work plans for the activities being carried out.

Design and development activities

Within the framework of design and development activities, graduates shall be able to:

- select appropriate hardware components, such as processors, memory, and input/output devices, in designing the architecture of computing systems, including their hardware and software components;
- design electronic circuits for computational processes;
- develop the hardware and software support of computing systems in the field of computer engineering based on improving energy efficiency, ensuring information security, and applying multimedia technologies;
- develop design and software documentation;

- apply international and professional standards of information technologies, modern methods, instrumental tools, and computing facilities in practice in accordance with the field of specialization.

Information and Analytical Activities

Within the framework of information and analytical activities, graduates shall be able to:

- assess the effectiveness of projects by analyzing project data and evaluating project performance;
- plan the use of production processes and resources, ensure information security, and analyze potential emergency risks;
- prepare reports based on the results of information and analytical activities, extract analytical information, and structure and systematize it;
- evaluate the effectiveness of managerial decisions.

Engineering and Entrepreneurial Activities

Within the framework of engineering and entrepreneurial activities, graduates shall be able to:

- develop start-up projects based on digital technologies in the field of computer engineering, as well as start and manage businesses;
- model, develop, and provide services related to multimedia technologies and resources;
- provide services for ensuring information security, secure data storage, and protection of information systems;
- analyze market opportunities for engineering products and develop innovative products or services.

Activities Related to the Application of Multimedia Technologies

Within the framework of activities related to the application of multimedia technologies, graduates shall be able to:

- develop multimedia resources and organize the use of multimedia tools in the activities of enterprises, organizations, and institutions;

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- master computer graphics and animation, audio and video technologies, produce multimedia products, and provide virtual reality services;
- plan multimedia tools and resources required for implementing hardware and software production processes;
- develop and apply field-specific multimedia tools in scientific and practical research activities.

Activities Related to Ensuring Information Security of Computer Systems

Within the framework of activities related to ensuring information security of computer systems, graduates shall be able to:

- develop information security policies and implement measures to ensure information security in the activities of enterprises and organizations;
- apply methods and tools for ensuring information security in the design and production of hardware and software tools in computer engineering;
- provide services for implementing information security in computer systems, information networks, and data centers;
- carry out activities in the field of information services and press services, while ensuring information security.

Intelligent Data Analysis Activities

Within the framework of intelligent data analysis activities, graduates shall be able to:

- organize decision-making processes in the activities of enterprises, organizations, and institutions;
- design, produce, and apply artificial intelligence technologies, as well as related hardware and software tools;
- widely apply methods and tools of intelligent data analysis and decision-making in scientific research and production processes;
- automate business processes in economic sectors and industries, and introduce artificial intelligence technologies into decision-making and management activities;

- process and statistically analyze big data, and manage enterprise databases.

2. REQUIREMENTS FOR PROFESSIONAL COMPETENCIES

Graduates of the **60610300 — Computer Engineering** Bachelor's Degree Program must possess the following professional competencies:

- develop fundamental general professional knowledge, practical skills, and abilities necessary for studying and thoroughly mastering specialized disciplines within the field of study;
- gain an understanding of the main achievements, problems, and development prospects in the areas of professional activity corresponding to the field of study;
- acquire knowledge of digital technologies and the field of computer engineering, including their classification, essence, and main characteristics;
- understand the interrelation of computer engineering technologies with other functional areas;
- acquire knowledge of the functional areas of computer technologies, including digital data processing, parallel and distributed processing, computer architecture, digital device design, network technologies, embedded systems, robotics and automation, artificial intelligence, and machine learning engineering;
- conduct research on computer systems and make decisions by analyzing management issues in computer systems;
- analyze and design digital technologies;
- possess scientific knowledge, practical expertise, and skills in both theory and practice.

3. REQUIREMENTS FOR PRACTICAL TRAINING (INTERNSHIPS)

Practical training (internships) is intended to reinforce the theoretical knowledge acquired through compulsory and elective courses and integrate it with practical experience in computer engineering and information and communication technologies (ICT). The internships aim to develop professional competencies, technical skills, problem-solving abilities, teamwork, and practical experience required for computer engineering professionals.

The educational program includes the following types of internships:

1. Introductory Educational Internship

This internship is designed to familiarize students with the professional environment of computer engineering, ICT organizations, computer systems laboratories, data centers, research institutions, technology companies, and public and private enterprises. Students become acquainted with the structure and operation of computer systems, hardware and software components, network infrastructure, microprocessor-based systems, information security principles, and professional ethics.

2. Industrial (Professional) Internship

This internship enables students to apply theoretical knowledge in real computer engineering and ICT environments. During the internship, students participate in the design, configuration, integration, testing, and maintenance of computer systems, hardware and software components, computer networks, embedded systems, microprocessor-based systems, cybersecurity solutions, data processing systems, and ICT infrastructure. Students gain practical experience by working individually and as members of engineering teams using modern computer engineering tools and technologies.

3. Pre-Graduation Internship

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The pre-graduation internship is conducted in ICT organizations, computer engineering enterprises, IT departments, data centers, research institutions, technology centers, or governmental organizations. It enables students to integrate the theoretical knowledge and practical skills acquired throughout the program, collect data and materials for their graduation project or thesis, conduct analysis, design, integration, and evaluation of computer engineering systems, and prepare for independent professional practice.

4. STRUCTURE OF THE MODULE CATALOGUE

The module catalogue is structured to ensure the systematic development of theoretical knowledge, professional competencies, practical skills in computer engineering, research abilities, and innovation capabilities. It consists of:

- General education courses;
- Core compulsory courses;
- Specialized professional courses;
- Elective courses;
- Practical training (internships);

No.	Module Code	Subject Name, Blocks, and Activity Types	Total Workload	Credits	Semester
		Compulsory modules			
1.01	UEY1104	The contemporary history of Uzbekistan	120	4	1
1.02	DIN1204	Religion Studies	120	4	2
1.03	FAL1204	Philosophy	120	4	2
1.04	XOT1108	Foreign language 1,2	240	8	1,2
1.05	HIS1112	Calculus 1,2	360	12	1,2
1.06	FIZ1110	Physics 1,2	300	10	1,2
1.07	DIT1406	Differential equations	180	6	4
1.08	CHA1306	Linear algebra	180	6	3
1.09	DAS1110	Programming 1,2	300	10	1,2

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1.10	AKY1104	Academic writing	120	4	1
1.11	MAB1306	Database	180	6	3
1.12	KIA1406	Cybersecurity fundamentals	180	6	4
1.13	MTA1204	Data structures and algorithms	120	4	2
1.14	EVS1306	Electronics and circuits 1	180	6	3
1.15	DIT1306	Discrete structures	180	6	3
1.16	KOT1306	Computer networks	180	6	3
1.17	SIA1406	Artificial intelligence fundamentals	180	6	4
1.18	EVS1406	Probability and statistics	180	6	4
1.19	KAT1506	Computer Organization	180	6	5
1.20	INL1704	Individual project	120	4	7
1.21	STI1506	Signal and Image Processing	180	6	5
1.22	KMA1606	Computer Architecture	180	6	6
1.23	ORT1506	Embedded Systems	180	6	5
1.24	OPTI 704	Operating Systems	120	4	7
1.25	GATI 704	Geoinformation Technologies	120	4	7
2.00		Elective modules	1800	60	4-7
		Qualification: Computer Systems Engineer			
		Total	7200	240	
		Internship	900	30	8
		Final State Attestation			
		Total	900	30	
		GRAND TOTAL	7200	240	

Bibliographic Information

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Keywords:

Professional activity type; field of study; object of professional activity; area of professional activity; core bachelor's curriculum and course syllabi; profile; learning outcomes; industries and sectors; qualification requirements; bachelor's educational process; quality assessment and control; independent learning; production; competence; organizational and managerial activity; professional internship; bachelor's graduation thesis; state attestation; academic course block; higher education institution; educational process; credit-module system; service provision; creative thinking; computer engineering; computer systems; hardware and software systems; information and communication technologies; project design and engineering; scientific research process.

The main types of higher education institutions and personnel consumers agreed upon by the developers

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